Wildfire Exposure and Farmland Values: Evidence from Manitoba

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Abstract

This paper examines whether wildfire exposure is capitalized into Manitoba farmland transaction values. Provincial sales records, 1987–2021, are linked to parcel geometry and Canadian wildfire records, giving 56,352 arm's-length sales of bare farmland. Exposure is measured as burned area within 10, 25 and 50 km over trailing windows of one to five years. Annual aggregate relationships between wildfire activity and median prices are not robust. In sale-level regressions, most of the raw negative association disappears once municipality fixed effects are added, but within municipality and year, farmland with more burned area nearby still sells for less (-0.036 on log burned area within 10 km). This association largely reflects the places where fires occur: soil capability, ecoregion and land cover account for about 79% of it, and burned area in the three years after a sale is associated with lower prices as strongly as burned area before it. A stacked event study around large nearby fires shows no clear post-fire change but rests on few fires and has limited power. The evidence provides no support for a price response to fire events, and the data cannot separate fire-risk capitalization from other characteristics of land at the forest fringe.

Keywords: wildfire; farmland values; capitalization; risk perception; event study; Manitoba

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1 Introduction

Farmland values provide a useful way to study how environmental risk is capitalized into agricultural asset markets. In standard capitalization models, the value of agricultural land reflects the discounted stream of expected future net returns from farming. This means that land prices should respond not only to current production conditions, but also to changes in expected profitability, risk, and the long-run attractiveness of farming and the land in a particular location. Climate-related shocks are therefore likely to matter for farmland values when they alter either the productive capacity of land or the way buyers and sellers perceive the risks associated with owning and operating that land.

This paper examines whether wildfire exposure is capitalized into Manitoba farmland transaction values. The question is important because wildfire is an increasingly salient climate-related risk across many agricultural regions, yet its effect on farmland markets is not straightforward. Unlike drought, heat, or flooding, wildfire rarely damages the parcel being sold directly. Instead, nearby wildfire activity may affect farmland values through more indirect channels. The primary mechanism emphasized in this paper is risk perception. A nearby wildfire can make future fire risk more salient to buyers, even when the land itself remains usable for production. In that setting, wildfire exposure may reduce willingness to pay by changing beliefs about the safety, reliability, or long-term desirability of the land in the area.

Two secondary mechanisms are also relevant. First, wildfire may weaken local economic activity in burned or nearby areas. If fires disrupt local infrastructure, reduce nearby population or business activity, or create temporary declines in local income and services, those effects may spill over into farmland markets. Second, wildfire exposure may affect expected ownership costs through insurance and risk-management channels. Even if farmland remains productive, higher expected insurance costs, more limited insurance availability, or greater perceived difficulty in protecting farm assets may be capitalized into transaction prices. Smoke is a further possible channel, although its effect on specific crops is ambiguous

in sign. These mechanisms imply that the relevant wildfire exposure measure is unlikely to be a simple contemporaneous indicator of whether a parcel burned. Instead, the effects may operate through nearby cumulative exposure over a medium-run period, where recent fires inform expectations about future risk and local market conditions.

The empirical analysis links Manitoba farmland sales records to parcel geometry and to wildfire records from the Canadian National Fire Database. A central data task is isolating genuine farmland transactions and locating them correctly. Most sales in the provincial records are residential or commercial, many recorded prices are nominal transfers, and sales of several parcels are recorded across several records. Restricting the data to arm's-length sales of bare farmland, with every parcel of each sale linked, yields 56,352 sales from 1987 to 2021. Wildfire exposure is measured as burned area within 10, 25, and 50 km of each sale over trailing windows of one to five years, and fire counts are used as a secondary measure. The main outcome is the logarithm of price per acre.

A descriptive wildfire history figure (Figure 1) also motivates the empirical design. Fire counts and area burned capture different dimensions of wildfire activity in Manitoba, and the historical series is shaped both by changes in fire detection and by a small number of very large burned-area years. This motivates the paper's emphasis on burned area and event-based local comparisons rather than treating annual fire counts as the main treatment variable.

The empirical strategy is intentionally sequential. The paper begins with annual aggregate evidence relating Manitoba-wide wildfire activity to annual median farmland prices, which provides descriptive context but not identification. It then moves to sale-level regressions that progressively add year and municipality fixed effects, in order to distinguish broad spatial differences from more localized wildfire variation, and examines how the association varies with distance and time window. Because a within-municipality association remains, the next step asks what lies behind it: whether it survives controls for the landscape in which sales take place, and whether its timing is consistent with a response to fire. The final stage uses stacked local event-study designs that compare areas experiencing a major nearby fire with

geographically proximate untreated areas.

The results support a cautious interpretation. The annual aggregate evidence does not show a stable relationship between wildfire activity and farmland values. In the sale-level regressions, large negative associations in specifications with limited controls shrink sharply once municipality fixed effects are added, indicating that most of the raw relationship reflects persistent differences across places. Within municipality and year, more burned area nearby is still associated with lower prices, and the association weakens with distance: the coefficient on log burned area in the prior three years is -0.036 at 10 km, -0.020 at 25 km, and -0.014 at 50 km. Exposed sales, however, lie in more forested landscapes, on poorer soils, and mostly outside the Prairies ecozone. Soil capability, ecoregion and land cover account for about 79% of the 10 km association, burned area in the three years after a sale is associated with lower prices about as strongly as burned area before it, and fire 6 to 20 years earlier is associated with lower prices too. The event-study designs show no clear post-fire change in prices, but they rest on few fires, fail a joint pre-trend test, and could not detect a change of the size of the cross-sectional association. The data provide no evidence of a price response to fire events, but cannot separate the pricing of persistent fire risk from other features of land at the forest fringe.

The contribution of this paper is threefold. First, it extends the farmland value literature by examining wildfire as a climate-related risk that may be capitalized into agricultural land markets through expectations, reduced local economic activity, and expected ownership costs, rather than through direct productivity losses. Most hedonic evidence on wildfire comes from residential housing markets in the western United States (Ma et al., 2024; McCoy & Walsh, 2018; Mueller et al., 2009; Stetler et al., 2010); this paper adds evidence from Canadian farmland. Second, it builds a parcel-linked framework for measuring wildfire exposure around individual farmland transactions and shows that the composition of administrative sales records matters: most linked records are not farmland sales, many prices are nominal, and sales of several parcels must be assembled from several records before acreage and location

can be measured. Third, it shows that within-municipality associations between nearby fire and land prices largely reflect the characteristics of the places where fires occur, using the landscape and the timing of fires relative to sales, and it documents the limits of event-based designs in sparse rural land markets.

2 Related Literature

2.1 Farmland values and the capitalization of expected returns

Much of the literature on farmland prices begins from the idea that agricultural land values reflect the present value of expected future net returns. In its simplest form, the capitalization model implies that farmland prices depend on the stream of expected returns to agriculture and the rate at which those returns are discounted into the present (Brueckner, 1990; Burt, 1986; Chambers, 1924; Nickerson & Zhang, 2014). This basic framework remains useful because it provides a direct link between environmental conditions and land transaction values. If climate affects the expected profitability, risk, or future use of agricultural land, those effects may be capitalized into observed farmland prices.

Early work in this area emphasized the role of farming returns in shaping land values. Chambers (1924), for example, used historic cash rents and mortgage interest rates to infer the expected increase in returns to farmland and concluded that farmland markets appeared to base expectations on recent experience. Later studies refined this basic intuition. Alston (1986) found that changes in real land price growth could be explained by real growth in net rental income to land, while Burt (1986) provided econometric support for the capitalization formula in a model linking farmland rents to farmland prices. Together, these studies established the broad proposition that farmland values are forward-looking and that expectations about future returns are central to understanding price movements.

At the same time, later work showed that the relationship between observed land prices and current farm income is not always straightforward. Studies of farmland booms and

busts, particularly those focused on the 1970s and 1980s, noted that observed land values often move more sharply than rents or realized farm returns alone would suggest (Falk, 1991; Falk & Lee, 1998; Just & Miranowski, 1993). This strand of literature does not reject the capitalization framework so much as it emphasizes that expectations, inflation, credit conditions, and alternative uses of land may all affect the mapping from current farm returns to observed farmland prices. For present purposes, the key implication is that climate-related effects on land values are likely to operate through expectations about future returns rather than through current farm income alone.

2.2 Proxies for expected returns to farming

A central empirical issue in the farmland literature is how expected returns to farming should be measured. Because expectations are not directly observed, economists have relied on a variety of proxies. Some studies use cash rents or rental income as a direct measure of returns to land (Burt, 1986; Chambers, 1924). Others use broader measures such as net farm income, net crop returns, farm cash receipts, or combinations of agricultural receipts and production costs (Goodwin et al., 2011; Nickerson & Zhang, 2014). Still others model expectations more explicitly, using lagged values, moving averages, or adaptive-expectations structures (Veeman et al., 1993; Weisensel et al., 1988).

The choice among these proxies matters because each one captures a somewhat different aspect of agricultural profitability. Goodwin et al. (2011), for example, use a five-year average of net market returns based on the difference between agricultural sales receipts and production costs. Just and Miranowski (1993) compare several approaches to modeling expectations, including rational, adaptive, extrapolative, and naïve expectations, and conclude that the naïve specification provides the best empirical fit in their setting. In Canadian work, Goodwin and Ortalo-Magné (1992) model expected values of government support, yields, and producer prices using realized values, while Weisensel et al. (1988) and Veeman et al. (1993) incorporate adaptive expectations into Prairie farmland price analysis. What emerges from

this literature is not a single preferred proxy for expected returns, but rather a recognition that researchers must choose a measure consistent with the economic mechanism under study.

This issue is especially important in a climate-related context, because climate can affect yields, input use, crop choice, production timing, output variability, and farming costs simultaneously, and is therefore unlikely to be fully captured by any one simple farm-income measure.

2.3 Climate, farmer decision-making, and adaptation

The literature on climate and agriculture adds an important further insight which is that climate affects agricultural outcomes not only directly, but also through the decisions farmers make in response to changing conditions. This point is especially clear in the contrast between production-function approaches and Ricardian approaches. Production-function studies estimate the effect of climate on output while holding production practices fixed. By contrast, the Ricardian approach infers climate impacts from farmland values and therefore allows, at least in principle, for adaptation in crop choice, input use, and management practices (Mendelsohn et al., 1994, 1996).

This distinction matters because farmers do not simply experience climate, they respond to it. Changes in temperature, precipitation, and other environmental conditions may alter which crops are grown, when production activities occur, how intensively inputs are used, and how attractive a parcel of land is as a long-term investment. In that sense, farmland values can be viewed as incorporating the consequences of both climatic conditions and farmer adaptation. Mendelsohn et al. (1994) argue explicitly that farmers are expected to adjust rather than remain locked into a single production practice as climate changes. Later work by Ortiz-Bobea and Just (2013) similarly emphasizes that modelling adaptation is central to any meaningful assessment of climate impacts in agriculture.

This is a major reason why farmland transaction values are an informative outcome: whereas annual production data capture short-run effects, land values summarize longer-

run expectations about profitability, risk, and the agricultural and non-agricultural choices available to the operator.

2.4 Climate and farmland values in the Ricardian tradition

The best-known approach linking climate directly to farmland values is the Ricardian method developed by Mendelsohn, Nordhaus, and Shaw (1994; 1996). The key idea is that if climatic conditions affect agricultural returns, then those climatic conditions should be reflected in farmland prices. Using cross-sectional regressions of farmland values on climate variables, Mendelsohn et al. (1994) show that some temperature and precipitation measures are associated with lower land values while others are associated with higher land values. They then use these coefficients to project the effects of future climate change on farmland values. An important conclusion from this work is that estimated damages are smaller under the Ricardian approach than under fixed-practice production-function approaches because the Ricardian framework implicitly incorporates adaptation (Mendelsohn et al., 1994, 1996).

Subsequent work extended and refined this approach. Reinsborough (2003) applied a Ricardian model to Canadian farmland and found that the expected effects of climate change were relatively modest. Weber and Hauer (2003), also using Canadian data, argued that earlier estimates had been overly pessimistic and reported climate-related gains in farmland values across regions. Mendelsohn and Reinsborough (2007) later extended the Ricardian framework to both the United States and Canada. More recent contributions have focused on methodological refinement. Massetti and Mendelsohn (2011) show how Ricardian models can be estimated using panel data, while Ortiz-Bobea (2020) argues that cash rents may sometimes be a preferable dependent variable because they are less likely than farmland prices to reflect omitted non-agricultural influences such as urban pressure.

This literature has also generated important critiques. Severen et al. (2018) argue that conventional Ricardian estimates may be biased if current land prices already embody beliefs about future climate change, and they propose a forward-looking Ricardian approach that

accounts for market expectations more explicitly. This critique is useful because it reinforces the broader point that farmland values are not simply functions of current climate, but of how market participants interpret the long-run implications of climatic conditions for agricultural profitability. A related concern applies to any cross-sectional estimate: places that differ in climate, or in exposure to a hazard, also differ in other characteristics that affect land values.

2.5 Other determinants of farmland values and their relevance to climate analysis

Although climate may affect expected returns to farming, the farmland literature makes clear that it is not the only determinant of land values. Interest rates and discount rates remain fundamental because they determine how strongly future returns are capitalized into current land prices (Alston, 1986; Just & Miranowski, 1993). Inflation may also matter through its interaction with real discount rates and the role of farmland as a hedge against inflation (Baker et al., 2014; Just & Miranowski, 1993). In addition, the non-farm value of land can have substantial effects on transaction prices, especially in areas subject to urban development pressure (Livanis et al., 2006; Nickerson & Zhang, 2014; Plantinga et al., 2002).

This point is particularly important for climate analysis. If farmland values reflect strong development premiums, amenity values, or other non-agricultural influences, then the agricultural effect of climate may be harder to isolate. Conversely, in regions where land values remain closely tied to farming returns, climate-related changes in expected agricultural profitability may be more readily capitalized into observed prices. For this reason, the institutional and geographic context of the land market matters a great deal. In Prairie settings, where agricultural returns remain a comparatively important component of land value, farmland transaction data may provide a particularly useful lens through which to study how environmental conditions influence longer-run economic outcomes.

2.6 Wildfire, natural hazards, and property values

A separate hedonic literature estimates how wildfire and other natural hazards are capitalized into property values. Most of this work studies residential housing in the western United States. Mueller et al. (2009) find that repeated wildfires in southern California reduce house prices, with larger effects after a second fire, consistent with buyers updating beliefs about local hazard. Stetler et al. (2010) find large and persistent negative effects of proximity to and views of burned areas on home values in northwest Montana. McCoy and Walsh (2018) show that prices of high-risk homes on the Colorado Front Range fall after nearby fires but recover within one to two years, which they interpret as a temporary increase in risk salience. Ma et al. (2024) study how the disclosure of wildfire hazard zones in California affects home prices, and Dong (2024) examines the impact of California wildfires on home values. Together, these studies suggest that wildfire can be capitalized through information and salience even when properties are not physically damaged, and that the effect may decay over a few years.

This literature also distinguishes the price of a persistent hazard from the response of prices to an event. In a fully informed market, a known hazard is capitalized into land values whether or not it has recently materialized, and an event changes prices only to the extent that it changes beliefs about future risk. Donovan et al. (2007) find that house prices in Colorado Springs were positively correlated with wildfire risk until the city published parcel-level risk ratings, after which the correlation disappeared. Bin and Landry (2013) find that flood-zone discounts in North Carolina appeared only after major hurricanes and faded within about five to six years, and the recovery documented by McCoy and Walsh (2018) points the same way. A cross-sectional discount for land in hazardous places is therefore consistent with risk capitalization but also with any characteristic of those places that lowers land values, while a change in prices after an event isolates the salience response.

Evidence for agricultural land is much thinner. Annan and Bigelow (2025) use a difference-in-differences design on Oregon agricultural land transactions from 2000 to 2023 and report preliminary estimates of large declines in value for parcels within 2 km of a wildfire. Related

work on other hazards shows that farmland markets do price natural-hazard risk: Wang (2021) uses a boundary discontinuity design to estimate the capitalization of flood risk into farmland values in Lancaster County, Pennsylvania.

The mechanisms considered in this paper also connect to work outside the property-value literature. Wildfire smoke can affect crop yields in both directions: Behrer and Wang (2024) find that low-density smoke plumes slightly raise Midwest corn and soybean yields, likely by increasing diffuse light, while high-density plumes reduce them. Large fires can depress local employment and income for several years (Coulombe & Rao, 2025; Kabore et al., 2023), and higher expected insurance costs can be capitalized into property values (Hennighausen et al., 2023; Keys & Mulder, 2024). These channels are discussed further in Sections 4 and 6.

2.7 Implications for the present study

Taken together, the literature suggests that climate can affect farmland transaction values through its influence on expected returns to farming, but that the empirical measurement of those expected returns remains unsettled. Researchers have used rents, net income, receipts, production costs, moving averages, and adaptive-expectations structures to proxy the stream of returns capitalized into land values (Burt, 1986; Chambers, 1924; Goodwin et al., 2011; Just & Miranowski, 1993). At the same time, the climate literature indicates that land markets may provide a useful way to observe the combined effects of environmental conditions and farmer adaptation (Mendelsohn et al., 1994, 1996; Ortiz-Bobea & Just, 2013; Reinsborough, 2003).

The main implication for this research is that climate-related hazards should not be treated simply as exogenous environmental characteristics attached to land. They shape the expected profitability and risk of farming, and farmland transaction values capture how market participants assess these longer-run consequences. The wildfire hedonic literature adds that such hazards can be capitalized through beliefs and salience rather than physical damage alone, and that the relevant effects may be local and may fade over several years.

The distinction between persistent risk and events, and the cross-sectional critique, imply that associations between hazard exposure and land values must be checked against the characteristics of the places exposed. These considerations guide the exposure radii, time windows and diagnostics in the empirical analysis below.

3 Data

3.1 Sales records and parcel linkage

The sales records come from the Manitoba government's Property Assessment Services database (Province of Manitoba, n.d.-a) in two files. The first covers 1984 to 2016 and records, for each sale, the transaction date, municipality, an assessment identifier, the sale price and the property class assigned by the assessor (for example, farm bare land, farm land and buildings, residential, or industrial, commercial and institutional). The second covers 2017 to early 2021, contains bare farmland sales only, and records the same information without a property class. Parcel boundaries come from the provincial roll entry layer (Manitoba Land Initiative, 2018; Province of Manitoba, n.d.-b).

The identifier in the historical file is the assessment system's internal roll identifier, the key in the address of each parcel's public assessment report, and historical sales are linked to parcels by matching it within municipality. Each parcel of a sale is a separate record with the same sale identifier. Of 390,046 historical sales, 386,794 (99.2%) link on all of their parcels, 942 (0.24%) on some, and 2,310 (0.59%) on none; partially linked and unlinked sales are excluded. In the modern file a sale's first record carries its date and price, and any following undated records list further parcels of the same sale: 10,247 sale records and 8,795 further-parcel records (3,859 sales list more than one parcel). Every parcel is linked by roll number within its municipality, and a sale is kept only if all of its parcels link: 10,100 sale records (98.6%) link in full, 40 on some parcels and 107 on none. In both files, sale records that repeat an earlier record exactly (same date, price, municipality and roll numbers) are

dropped, keeping one of each: 287 in the modern file and 0 in the historical file.

Within municipality and year, the log of the 2026 assessed value of the linked parcels explains 0.54 of the variation in log sale price (R^2), and the log of linked acreage explains 0.33 (Appendix Table A10). Acreage is the sum of the linked parcel polygon areas,

$$A_i = \sum_{p \in i} \frac{Area_p}{4046.8564224}, \tag{1}$$

where $Area_p$ is parcel area in square metres, and the outcome is price per acre, $PPA_i = C_i/A_i$, where C_i is the sale price.

3.2 Construction of the farmland sample

Most linked sales are not farmland transactions. Table 1 shows how the farmland sample is constructed. Of the linked historical sales, 52,488 are classed as farm bare land; most of the rest are residential or commercial. The sample is restricted to sales of farm bare land, with a price of at least $1,000 (removing nominal transfers, which make up a large share of the modern file), linked parcels totalling at least 10 acres, and price per acre between the 1st and 99th percentiles of its sale year, from 1987 onward. The resulting sample contains 56,352 sales: 49,879 from 1987 to 2016 and 6,473 from 2017 to 2021, in 123 municipalities. 35% of modern sales in the sample cover more than one parcel.

Table 2 reports summary statistics. The typical sale is a quarter section of about 160 acres. The median price per acre is $373 in the historical period and $1,608 in the modern period, reflecting the appreciation of Manitoba farmland. About 10% of sales have any burned area within 10 km in the three years before the sale, 29% within 25 km and 57% within 50 km. Exposure within 10 km is mostly to small fires: the median burned area among exposed sales is 18 ha. Figure 2 maps the sales, the fires and forest cover.

The modern file has no property-class field, so each modern sale was checked against its first parcel's public assessment report, which records tax class and building value by tax

year. Reports were retrieved for 3,132 of the 6,473 modern sales (48%); 2,695 of these (86%) are confirmed as bare farmland at the sale's tax year, and 437 have buildings, a residential portion or a non-farm class. Excluding them leaves the main 10 km coefficient at -0.036 (Appendix Table A1).

3.3 Wildfire data and exposure measures

The wildfire data are fire point records from the Canadian National Fire Database, distributed through the Canadian Wildland Fire Information System (Natural Resources Canada, n.d.). Each record gives the reporting agency, ignition point, start date, out date where recorded, burned area and cause. The analysis uses 142,617 records from 23 agencies within 1,000 km of the sales, so fires in neighbouring provinces are included; fires in the United States are not. Prescribed burns, recorded with a separate cause code chiefly by Parks Canada, are planned, managed fires and are removed: 132 records with 97,575 ha of burned area. Records without a start date cannot be placed in time and are dropped: 3,279 records with 2,205,097 ha of burned area (3.3% of the area retained), of which 2,857 are Manitoba-agency records from 1996 to 2004. Section 5.7 reports results with these fires dated 1 July of their fire year. The Manitoba agency records cause only as human (51% of its records), lightning (48%) or unknown (21 records), so agricultural burning cannot be separated from other human-caused fires. As a robustness check, burned area is also measured from the fire perimeters of the National Burned Area Composite (Hall et al., 2020).

Wildfire exposure is measured in two ways. First, annual Manitoba series of area burned and fire counts, from Manitoba-agency records, are summed over trailing 1-, 3-, 5- and 7-year windows and used in exploratory annual regressions. Second, exposure is measured for each sale. A fire counts toward a sale's exposure only if it ended before the sale date: where the record has an out date (81% of records), that date is used; otherwise the fire counts only if it started by the last day of the month before the sale month. A sale's exposure BA_{irw} is the total burned area (ha) of fires whose ignition point lies within radius $r \in \{10, 25, 50\}$ km of the

sale and whose date falls within the trailing window of $w \in \{1, 3, 5\}$ years; the corresponding fire counts are a secondary measure. The distinction between counts and burned area matters because the historical fire-count series is affected by changes in detection and reporting over time (Figure 1).

3.4 Land cover, soil capability and ecoregion

Land cover comes from the Agriculture and Agri-Food Canada Land Use maps (Agriculture and Agri-Food Canada, n.d.-b) for 2000, 2005, 2010, 2015 and 2020; each sale uses the latest map at or before its sale year, and the 24,191 sales from 1987 to 1999 use the 2000 map, which postdates the sale. Three measures are constructed: forest share within 10 km, distance to the nearest 300 m cell that is at least half forest (median 1.3 km), and grassland and shrubland share within 10 km, a partial proxy for pasture. Soil capability comes from the Canada Land Inventory's land capability for agriculture (Agriculture and Agri-Food Canada, n.d.-a) and ecoregions from the National Ecological Framework (Ecological Stratification Working Group, 1995).

3.5 Consistency of the two sales files

The change of sales file at 2017 does not shift prices. Within the same municipalities, quarter-section prices from 2014 to 2019 change at 2017 by -0.089 log points beyond a linear trend (standard error 0.064, $p = 0.17$), and by -0.115 ($p = 0.13$) using only modern sales confirmed as bare land. Relative to the Statistics Canada estimate of the value of farmland and buildings per acre, the median sale price per acre averages 0.66 in the historical years and 0.70 in 2017–2020. Year fixed effects absorb any remaining difference in level between the files.

4 Empirical Strategy

The historical wildfire series provides an important starting point. Figure 1 shows that the number of recorded fires in Manitoba rises steeply through the 1960s and remains elevated, which likely reflects improvements in detection and reporting as well as changes in fire activity. Fire counts are therefore a noisy measure of economically relevant exposure, and burned area receives greater emphasis. Area burned, in turn, is dominated by a few very large fire years, notably 1989 and 1994, which weakens the case for province-wide annual regressions and strengthens the case for treating major fires as local shocks.

The mechanisms discussed in the introduction differ in their likely reach and timing. Risk perception is most plausibly local and strongest in the years immediately after a fire, since salience effects in housing markets fade within one to two years (McCoy & Walsh, 2018). Reduced local economic activity may extend over a wider area and persist for several years (Coulombe & Rao, 2025; Kabore et al., 2023). Insurance and ownership costs may be capitalized gradually as expectations adjust (Hennighausen et al., 2023; Keys & Mulder, 2024), although bare farmland has few insurable structures. Smoke travels well beyond the radii considered here and its effects on specific crops is ambiguous in sign (Behrer & Wang, 2024). Because these mechanisms differ in scale, the analysis does not commit to a single radius; it reports burned area within 10, 25 and 50 km over one to five years and interprets the pattern across them. These mechanisms are not the only explanations for an association between nearby fire and farmland prices: land close to fire-prone, forested areas may differ systematically from other farmland, for example in soil quality or land use. The design addresses this possibility directly in its third stage.

4.1 Annual aggregate analysis

The first stage uses annual aggregate data to examine whether broad temporal relationships between farmland values and wildfire activity are visible. The annual series covers 1987 to

2020 and has only 34 observations, so this stage is treated strictly as an exploratory diagnostic. For each wildfire measure and window, three specifications are estimated: a level log-log model,

$$\ln(Y_t) = \alpha + \beta \ln\Big(1 + W_t^{(w)}\Big) + \phi M_t + u_t, \tag{2}$$

where Y_t is the median price per acre in year t, $W_t^{(w)}$ is Manitoba area burned or fire count summed over the trailing w years, and M_t indicates the 2017–2021 sales file; a model that adds a linear time trend γt; and a first-difference model,

$$\Delta \ln(Y_t) = \alpha + \beta \, \Delta \ln\Big(1 + W_t^{(w)}\Big) + \phi \, \Delta M_t + u_t. \tag{3}$$

4.2 Sale-level regressions

The main analysis is at the sale level. Let $y_i = \ln(PPA_i)$ and $a_i = \ln(A_i)$. Wildfire exposure enters in logged form as $\ln(1 + E_i)$. The baseline specification is

$$y_i = \alpha + \beta \ln(1 + E_i) + \delta a_i + \varepsilon_i, \tag{4}$$

a cross-sectional benchmark. A second specification adds year fixed effects λ_t, which absorb province-wide price movements, and a third adds municipality fixed effects μ_m,

$$y_i = \alpha + \beta \ln(1 + E_i) + \delta a_i + \lambda_t + \mu_m + \varepsilon_i. \tag{5}$$

The linkage tier (historical or modern) is determined by the sale year, so a tier indicator is absorbed by the year effects. If wildfire coefficients attenuate substantially once municipality effects are added, much of the raw relationship reflects persistent differences across places rather than local exposure within municipalities. This progression is estimated first for burned area and fire counts within 25 km. Equation (5) is then estimated for burned area at each radius $r \in \{10, 25, 50\}$ km and window $w \in \{1, 3, 5\}$ years, for all sales and for the

historical and modern periods separately. Because exposure is spatially correlated, main tables report Conley (1999) standard errors that allow correlation between sales within 25 km of each other; the appendix reports standard errors clustered by municipality.

4.3 What lies behind the within-municipality association

If a within-municipality association remains, the next step is to ask whether it reflects a response to fire or the characteristics of the places where fire occurs. Two sets of checks follow from the sale-level results. The first compares exposed and unexposed sales on the landscape, and then re-estimates equation (5) adding, in turn, soil capability and ecoregion; land cover (forest share, distance to forest and grassland share); and burned area 6 to 20 years before the sale. If the association is a response to recent fire, it should survive controls for the landscape.

The second set uses the timing of fires relative to the sale. The model adds burned area within the same radius in the three years after the sale,

$$y_i = \beta \ln(1 + BA_{ir3}) + \gamma \ln(1 + Lead_{ir3}) + \delta a_i + \lambda_t + \mu_m + \varepsilon_i, \tag{6}$$

where $Lead_{ir3}$ is the burned area of fires that started after the sale and within three years of it. Future fire cannot have affected the sale price, so under a fire-response interpretation γ should be close to zero and clearly smaller than β; if both reflect the propensity of the surrounding land to burn, γ should be similar to β. Because fire records end in 2021, this model uses the 53,025 sales on or before 31 December 2018. Burned area 6 to 20 years before the sale, which recent fire cannot have caused, measures the long-run fire regime of the location.

4.4 Stacked local event studies

The final stage asks whether prices in an area change after a large nearby fire, relative to nearby areas without one. Sales are assigned to 10 km hexagonal cells. A cell is treated in the first year G_c in which burned area within 25 km of its centroid reaches 500 ha, and each sale is placed in event time $k = t - G_c$, with sales in the event year before the threshold was crossed assigned to $k = -1$. For each treated cell, a stack is formed with control cells in a 25–50 km ring that are not treated during the event window (-3 to $+3$ years), and support rules require sales before and after the event in both treated and control cells. The stacked event-study model is

$$y_{ics} = \sum_{k \neq -1} \beta_k D_{ics}^{(k)} + \delta a_i + \mu_{cs} + \lambda_{ts} + \varepsilon_{ics}, \tag{7}$$

with stack-specific cell and year fixed effects, and a pooled version replaces the event-time indicators with a single post-fire indicator,

$$y_{ics} = \beta Post_{ics} + \delta a_i + \mu_{cs} + \lambda_{ts} + \varepsilon_{ics}. \tag{8}$$

Constructing a separate stack for each treated event, with controls that are not yet treated during the event window, avoids the comparisons with already-treated units that make conventional two-way fixed-effects estimates difficult to interpret under staggered timing (Callaway & Sant'Anna, 2021; Sun & Abraham, 2021). Estimates are reported unweighted and with the corrective weights of Wing et al. (2024), and standard errors are two-way clustered by cell and stack (Abadie et al., 2023). As in the sequential design more broadly, broader designs are also estimated: a wider 50–100 km control ring, which addresses the possibility that nearby controls are partly treated (Butts, 2021); a first stacked design with a $\pm$5-year window and a 25–75 km ring; and a broad design with never-treated controls.

Three diagnostics accompany the event study. The pre-period coefficients are tested

jointly. Sensitivity of the post-period estimates to violations of parallel trends is assessed with the relative-magnitude bounds of Rambachan and Roth (2023). Because a pre-trend test with low power is uninformative (Roth, 2022), the minimum detectable effect of the pooled estimate at 80% power is reported and compared with the cross-sectional association implied by the event study's own treatment definition.

5 Results

5.1 Annual aggregate evidence

The annual aggregate analysis provides no robust evidence of a relationship between wildfire activity and farmland values. Table 3 reports the annual regressions. No area-burned coefficient is statistically significant in levels, with a trend, or in first differences (the smallest p-value is $p = 0.13$), and Figure 3 shows that year-to-year changes in the median price per acre are essentially unrelated to changes in trailing area burned. Fire counts behave differently: over 5- and 7-year windows their coefficients are negative and significant in the trend models (-0.232, $p = 0.001$; -0.291, $p < 0.001$) and in first differences (-0.124, $p = 0.035$; -0.140, $p = 0.024$). Because the recorded number of fires is shaped by changes in detection and reporting, these coefficients are hard to interpret as a price response. With so few observations and a series dominated by a few large fire years, the annual exercise is better treated as descriptive background than as a credible time-series identification strategy, and it motivates the shift to sale-level exposure measures.

5.2 Progressively added fixed effects

Table 4 reports the sale-level progression for burned area and fire counts within 25 km. In the baseline specification, all exposure measures are strongly associated with lower prices: the coefficient on log burned area is -0.137, -0.119 and -0.122 for the 1-, 3- and 5-year windows, and on log fire counts -0.454, -0.361 and -0.349. Adding year fixed effects reduces

these associations only modestly (to −0.101 for 3-year burned area). Once municipality fixed effects are added, the burned-area coefficients fall to −0.015, −0.020 and −0.021, and the fire-count coefficients to between −0.048 and −0.072. Most of the raw relationship between nearby fire and farmland prices therefore reflects persistent differences across municipalities: farmland in municipalities with more fire activity sells for less. A smaller association remains within municipalities and years.

5.3 Radius and window

Table 5 examines that remaining association across radii and windows. Within municipality and year, more burned area nearby is associated with lower prices at every radius and window, and the association weakens with distance: with the three-year window, the coefficient is −0.036 at 10 km, −0.020 at 25 km and −0.014 at 50 km, each $p < 0.001$ under Conley standard errors; at 10 km it is −0.029 for the one-year window and −0.038 for the five-year window. The pattern is the same for historical sales (−0.036 at 10 km), similar in size but less precisely estimated for the smaller set of modern sales (−0.031, $p = 0.098$), and holds with municipality-clustered standard errors (Appendix Table A2) and after a Holm adjustment for multiple testing (largest adjusted $p = 0.011$; Appendix Table A11). By the functional form alone, the 10 km coefficient implies that a sale with the median exposed burned area of 18 ha sells for about 10% less than an otherwise similar unexposed sale; the following sections ask whether this reflects fire.

5.4 Exposed and unexposed land, and landscape controls

Table 6 compares sales with any burned area within 10 km in the prior three years (5,549 sales) with all other sales. Exposed sales have an average forest share within 10 km of 0.44, against 0.16 for unexposed sales; they lie on average 0.6 km from forest, against 3.5 km; 40% of them are on land whose dominant soil class is 1 to 3, against 69%; and 35% lie in the Boreal Shield ecozone, against 1% of unexposed sales, while 20% lie in the Prairies, against 77%.

85% of exposed sales also had fire within 10 km 6 to 20 years before the sale, against 24%. Municipality fixed effects remove much of these differences but not all: within municipality and year, the difference between exposed and unexposed sales is 0.075 in forest share, −0.099 in the share on class 1–3 soils, and 0.19 in the share with fire 6–20 years earlier.

Table 7 adds the landscape controls in turn. At 10 km, soil capability and ecoregion reduce the coefficient from −0.036 to −0.020, and land cover reduces it further to −0.007 ($p = 0.14$): together they account for about 79% of the association. Adding burned area 6–20 years before the sale leaves it at −0.008. At 25 km the coefficient falls from −0.020 to −0.002, and at 50 km from −0.014 to −0.005. Among sales from 2000 onward, whose land-cover map predates the sale, a small controlled association remains (−0.012, $p = 0.031$; 32,158 sales), about a third of the uncontrolled association. The land-cover variables are themselves strongly associated with price: the coefficient on forest share within 10 km is −0.76 ($p < 0.001$). Farmland near forest sells for less, and that is where fires occur.

5.5 Timing of fire relative to the sale

Table 8 reports the lead placebo. Without landscape controls, burned area within 10 km in the three years after the sale is associated with lower prices (−0.035, $p < 0.001$) about as strongly as burned area in the three years before it (−0.030, $p < 0.001$), in the same model; at 25 km the lead coefficient (−0.020) is larger than the pre-sale coefficient (−0.016). With land cover, soil and ecoregion controls, the pre-sale coefficient becomes indistinguishable from zero (−0.005, $p = 0.26$) while the lead coefficient remains at least as large (−0.013; $p = 0.025$ under Conley standard errors and $p = 0.058$ with municipality clustering). Figure 4 shows how both coefficients change as the controls are added. Among sales from 2000 onward (columns 7–9; 28,837 sales, the sales from 2000 to 2018, whose lead window is fully observed), the pre-sale coefficient with controls is −0.011 ($p = 0.044$) and the lead −0.009 ($p = 0.13$): fire after the sale is associated with lower prices by a similar amount, but this smaller sample cannot establish the match with confidence.

Two land-use explanations fit an association with fire after the sale: the land sold lies in a fire-prone landscape, or buyers clear or burn land after purchase. The lead association comes from human-caused fires (-0.035, $p < 0.001$); lightning-caused fire after the sale is unrelated to price (0.001, $p = 0.90$). Excluding fires that ignite within 2 km of the sale leaves the lead coefficient at -0.033 ($p < 0.001$), so it does not come from burning on or next to the parcel sold (Appendix Table A5). Lightning-caused fires show no statistically significant association with price either before or after the sale: entered together, the pre-sale coefficient is -0.025 ($p = 0.073$) and the lead -0.014 ($p = 0.24$), and -0.002 and 0.004 with landscape controls; few sales are exposed to lightning-caused fires, so these estimates are imprecise. Table 10 splits pre-sale burned area by cause and size; it is descriptive, and the largest coefficients are on small human-caused fires under 10 ha (-0.044, $p = 0.037$).

Long-run fire history points the same way (Table 9). Burned area within 10 km 6 to 20 years before the sale is associated with lower prices (-0.040 alone; -0.037 alongside recent burned area), and including it reduces the coefficient on recent burned area from -0.036 to -0.025. Recent and long-past fire, and fire yet to come, are all associated with lower prices, the pattern expected if they all measure the fire-proneness of the land.

5.6 Event-study evidence

Table 11 and Figure 5 report the preferred stacked event study. It has 38 treated areas (stacks) with 527 treated sales, built around 13 distinct fires in 9 event years: each stack's event-year burned area comes mostly from one large fire, and several stacks share the same fire. The post-fire coefficients are imprecise. The pooled estimate is -0.029 (standard error 0.066), with a 95% confidence interval from -0.157 to 0.100; the average of the post-period coefficients relative to the year before the fire is 0.029 (standard error 0.062). The two differ because the pooled estimate compares the post period with all pre-period years, and the coefficient at $k = -3$ is large and positive ($+0.181$); dropping $k = -3$ gives a pooled estimate of 0.029 ($p = 0.67$). The Wing et al. weights leave the pooled estimate unchanged (-0.029),

and the wider control ring, the broader designs, the threshold and radius grid, and the leave-one-event-year-out runs are also imprecise and centred near zero (Appendix Table A8).

These estimates are inconclusive rather than evidence of no effect. The joint test of the pre-period coefficients rejects parallel pre-trends ($F = 3.59$, $p = 0.038$). The relative-magnitude bounds are wide: if post-period violations of parallel trends may be up to half the largest pre-period violation, the bounds for the average post-period effect run from -0.43 to 0.51. And the design has low power: its minimum detectable effect at 80% power is 0.184 log points, about 17%, while in the cross-section, sales with 500 ha or more burned within 25 km in the three years before the sale sell for -0.108 log points less ($p = 0.001$). The event study could not have detected a post-fire change of that size. This is the limit of event-based designs in a sparse rural land market: few large fires occur close to enough farmland sales to support local treated-control comparisons.

5.7 Robustness

The 10 km, three-year coefficient ranges from -0.038 to -0.032 across the main sample and the variants with buildings, with at least 40 acres and without trimming (Appendix Table A1). Measuring burned area from fire perimeters rather than ignition points gives -0.031 ($p < 0.001$) at 10 km, falling to -0.005 with landscape controls (Appendix Table A6). Controls from the 2026 assessment roll, which may be affected by exposure and are shown only as a check, reduce the coefficient to -0.012 (Appendix Table A9). Adding the fire records without a start date, dated 1 July, gives -0.036, and -0.006 with landscape controls (Appendix Table A7). In the event study, 4 of the 13 fires are recorded by Parks Canada in Riding Mountain National Park; excluding the 8 stacks within 10 km of the park leaves 30 stacks and a pooled estimate of -0.070 (standard error 0.081), and building the treatment with prescribed burns included gives 43 stacks and a pooled estimate of -0.001 (Appendix Table A8).

6 Discussion

The mechanisms that motivate the paper each imply a particular pattern. Risk perception implies that prices fall after a nearby fire, most strongly close to the fire and in the following years, and then perhaps recover as salience fades. Reduced local economic activity implies a post-fire decline over a wider area that persists for several years. Higher expected insurance or ownership costs imply a gradual decline after fires that change expectations, although bare farmland has few insurable structures. All three operate through fire events, and none implies that fire after a sale should be associated with lower prices. The evidence shows a within-municipality association that weakens with distance, as the risk-perception channel would suggest, but that association is largely explained by the landscape, is matched by fire after the sale and by fire long before it, and is not accompanied by a detectable post-fire change in the event study. The data therefore provide no evidence for any of the three channels operating through fire events, although the event study is too imprecise to rule out effects of the size of the cross-sectional association.

The landscape interpretation fits the evidence better. Farmland near fires lies near forest, on poorer soils and outside the Prairies ecozone, and it burned before and burns again after the sale. A persistent fire hazard could be capitalized into land values whether or not a fire has recently occurred, and the places that burn repeatedly are the places where such a hazard is highest. The lower prices of land near forest may partly reflect that hazard. They may equally reflect lower soil productivity, a greater share of forage and pasture, smaller and more irregular fields, distance from markets, or recreational demand. Fire risk and these characteristics are both driven by proximity to forest, and in this setting they cannot be separated. The association with small human-caused fires, many of which are likely agricultural or grass fires tied to land use, illustrates the same problem. The appropriate conclusion is that the data provide no evidence of a price response to fire events (with landscape controls, the association at the median exposure lies between about 5.0% lower and 0.7% higher), but cannot separate the capitalization of fire risk from other characteristics of land at the forest fringe.

The comparison with Annan and Bigelow (2025) is limited by differences in setting and measurement. Their effects are concentrated among Oregon parcels within 2 km of a fire, a scale at which fire is more likely to affect the parcel itself. The exposure measures here are defined over 10 to 50 km, around ignition points rather than perimeters, in a landscape where farmland meets the boreal forest along a long fringe. The results are therefore not directly comparable; they suggest only that, at the radii that sparse Manitoba sales allow, proximity to fire largely measures where land lies.

Because local fire exposure is so closely tied to the landscape, wildfire's broader reach through smoke is the natural next step. Smoke exposure at a farm in a given year depends on wind and on fire activity hundreds of kilometres away, so its year-to-year variation is much less tied to the local landscape than proximity to fire is. That variation identifies shorter-run responses: for crop yields and farm returns, which respond to smoke in the year it occurs and in both directions (Behrer & Wang, 2024), it is well suited, while for land values it identifies transitory or salience responses rather than the capitalization of long-run exposure. Persistent smoke exposure is itself geographic, since some regions lie downwind of fire-prone forest year after year, and a cross-section of average exposure would face the same confounding as fire proximity.

7 Limitations

Several limitations qualify the results. First, fires in the United States are not included in the exposure measures, so exposure is understated for sales near the southern border; because the forest fringe and most fire activity near the sales lie to the north, this is unlikely to drive the landscape pattern. Second, land cover for sales from 1987 to 1999 is measured with the 2000 map, after the sale, and the controlled association that remains among later sales is small but not zero. Third, the event study fails a joint pre-trend test and is underpowered for effects of the size of the cross-sectional association. Fourth, assessment reports were retrieved for about

half of modern sales, so some modern sales with buildings remain; excluding those identified leaves the results unchanged. Fifth, the Manitoba agency records cause only as human, lightning or unknown, so agricultural burning cannot be separated from other human-caused fires, and the lightning-caused classes rest on few exposed sales. Sixth, fire records without a start date cannot be timed; dating them to 1 July leaves the results similar.

8 Conclusion

This paper examines whether wildfire exposure is capitalized into Manitoba farmland transaction values. The central premise is that farmland prices reflect expectations about future returns, risk, and the long-run desirability of owning agricultural land in a particular location, so that wildfire may affect transaction values even when it does not burn the parcel being sold, most plausibly through risk perception and secondarily through local economic activity and expected ownership costs.

The evidence supports a cautious interpretation. The annual aggregate analysis does not reveal a robust relationship between province-wide wildfire activity and median farmland prices. In the sale-level regressions, most of the raw negative association between nearby fire and prices disappears once municipality fixed effects are added. A within-municipality association remains and weakens with distance, but it largely reflects the characteristics of the places where fires occur: it is explained to a large extent by land cover, soil capability and ecoregion, and it is matched by fire after the sale and by fire long before it. The event-study designs show no clear post-fire change in prices, but rest on few fires and have limited power. The data therefore provide no evidence of a price response to fire events and cannot separate the capitalization of persistent fire risk from other characteristics of land at the forest fringe.

Local fire proximity is a weak basis for identifying the effect of wildfire on land values because it is tied to the landscape. Smoke from distant fires, whose year-to-year variation is much less tied to local land characteristics, is the natural extension, beginning with its

effects on yields and farm returns.

Data Availability

Farmland transaction records are from the Province of Manitoba's Property Assessment Services sales database, and parcel boundaries are from the provincial roll entry and cadastral layers (Manitoba Land Initiative, 2018; Province of Manitoba, n.d.-a, n.d.-b). Wildfire records are from the Canadian National Fire Database, distributed through the Canadian Wildland Fire Information System under an end-user agreement (Natural Resources Canada, n.d.), and fire perimeters from the National Burned Area Composite (Hall et al., 2020). Land use, soil capability and ecoregion data are from Agriculture and Agri-Food Canada (Agriculture and Agri-Food Canada, n.d.-a, n.d.-b; Ecological Stratification Working Group, 1995). The R and Python code used to link the records, construct the sample and exposure measures, estimate all models and generate every number, table and figure in this paper is available from the author on request, together with the derived analysis files to the extent permitted by the data providers' terms of use.

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Table 1. Construction of the Farmland Sample

Step	Historical 1984–2016	Modern 2017–2021	Total
Records in source files		19,042	
Sale records in source files	390,046	10,247	400,293
Not an exact duplicate of an earlier sale record	390,046	9,960	400,006
Linked to parcels (every parcel of the sale)	386,794	9,819	396,613
Share of sale records linked	99.2%	98.6%	
Property class is farm bare land	52,488	9,819	62,307
Sale price at least $1,000	51,597	6,687	58,284
Acreage at least 10 acres	50,929	6,611	57,540
Price per acre trimmed (1st–99th pct. by year); 1987 onward	49,879	6,473	56,352

Notes: Steps are applied cumulatively. Historical sales come from the 1984–2016 provincial sales file, which records the property class of each sale and lists each of its parcels as a record with the sale identifier; parcels are linked on the assessment system's internal roll identifier within municipality. Modern sales come from the 2017–2021 bare farmland sales file, which has no property-class field; a sale's first record carries its date and price and any following undated records list further parcels of the same sale. Each parcel of a modern sale is linked by roll number within municipality. In both files a sale is kept only if every one of its parcels links, and acreage is the sum over its parcels (Section 3.1). Exact duplicate sale records (same date, price, municipality and roll numbers) are dropped, keeping one of each. Sources: `output/v4_corrected/sample/v4_sample_flow.csv`, `v4_modern_linkage_summary.csv`.

Table 2. Summary Statistics, Farmland Sample

	Historical 1987–2016	Modern 2017–2021	All 1987–2021
Sales	49,879	6,473	56,352
Municipalities	122	98	123
Median price per acre ($)	373	1,608	420
Mean price per acre ($)	637	2,192	816
Median acres	160	160	160
Mean acres	205	220	207
Share with burned area within 10 km, prior 3 years	0.101	0.078	0.098
Share with burned area within 25 km, prior 3 years	0.292	0.248	0.287
Share with burned area within 50 km, prior 3 years	0.571	0.583	0.572
Median burned area within 10 km if exposed (ha)	20.0	11.6	18.0
Mean forest share within 10 km	0.187	0.186	0.187
Median distance to forest (km)	1.34	1.34	1.34
Mean grassland/shrubland share within 10 km	0.021	0.020	0.021
Share with land cover measured after the sale	0.485	0.000	0.429

Notes: Exposure counts fires (Canadian National Fire Database points) that ended strictly before the sale date. Land cover from the AAFC Land Use maps (latest map at or before the sale year; sales before 2000 use the 2000 map). Source: `output/v4_corrected/phase5/p5_summary_stats.csv`.

Table 3. Annual Regressions of Median Price per Acre on Provincial Fire Activity

	Trailing window			
Model	1 year	3 years	5 years	7 years
Panel A. Manitoba area burned				
Level	−0.104	−0.185	−0.211	−0.205
	(0.067)	(0.119)	(0.165)	(0.194)
With linear trend	−0.001	0.043	0.010	−0.033
	(0.022)	(0.057)	(0.064)	(0.044)
First difference	−0.005	0.021	−0.005	−0.004
	(0.006)	(0.017)	(0.036)	(0.031)
Panel B. Manitoba fire count				
Level	−0.121	−0.562*	−0.390	−0.068
	(0.156)	(0.322)	(0.493)	(0.376)
With linear trend	0.013	−0.053	−0.232***	−0.291***
	(0.028)	(0.112)	(0.066)	(0.062)
First difference	−0.007	−0.005	−0.124**	−0.140**
	(0.011)	(0.056)	(0.056)	(0.059)

Notes: Dependent variable: log annual median price per acre (1987–2020; first differences 1988–2020). Regressor: $\ln(1+W)$, where W is area burned (thousand ha) or the number of fires over the trailing window. All models include an indicator for the 2017–2021 sales file (its change in first differences). Conventional standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Fire counts are sensitive to changes in detection and reporting over time and are shown for completeness. Source: `output/v4_corrected/main/annual/v4_annual_regressions_long.csv`.

Table 4. Sale-Level Regressions: Progressively Added Fixed Effects

	(1)	(2)	(3)
Panel A. Burned area within 25 km			
1-year window	−0.137***	−0.108***	−0.015***
	(0.020)	(0.021)	(0.005)
		[9,226 exposed sales]	
3-year window	−0.119***	−0.101***	−0.020***
	(0.019)	(0.019)	(0.005)
		[16,166 exposed sales]	
5-year window	−0.122***	−0.098***	−0.021***
	(0.019)	(0.018)	(0.006)
		[21,237 exposed sales]	
Panel B. Number of fires within 25 km			
1-year window	−0.454***	−0.365***	−0.048**
	(0.076)	(0.071)	(0.023)
		[9,226 exposed sales]	
3-year window	−0.361***	−0.295***	−0.069***
	(0.061)	(0.057)	(0.023)
		[16,166 exposed sales]	
5-year window	−0.349***	−0.269***	−0.072***
	(0.055)	(0.051)	(0.024)
		[21,237 exposed sales]	
Sales		56,346	
Year fixed effects		Yes	Yes
Municipality fixed effects			Yes

Notes: Dependent variable: log price per acre; all models control for log acreage. Exposure enters as ln(1+burned area, ha) in Panel A and ln(1+number of fires) in Panel B, counting fires within 25 km that ended before the sale date during the trailing window. Each cell is a separate regression. Conley standard errors with a 25 km cutoff in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (Conley). Exposed sales in brackets. Source: `output/v4_corrected/sale_level/v4_sale_level_main_table.csv`.

Table 5. Burned Area and Log Price per Acre, by Radius and Window

	Trailing window		
Radius	1 year	3 years	5 years
Panel A. All sales, 1987–2021			
10 km	−0.029***	−0.036***	−0.038***
	(0.009)	(0.009)	(0.010)
	[2,681]	[5,549]	[7,860]
25 km	−0.015***	−0.020***	−0.021***
	(0.005)	(0.005)	(0.006)
	[9,226]	[16,166]	[21,237]
50 km	−0.008**	−0.014***	−0.018***
	(0.003)	(0.004)	(0.004)
	[22,015]	[32,257]	[38,344]
Sales		56,346	
Panel B. Historical sales, 1987–2016			
10 km	−0.028***	−0.036***	−0.039***
	(0.009)	(0.010)	(0.010)
	[2,444]	[5,041]	[7,268]
25 km	−0.015***	−0.020***	−0.021***
	(0.005)	(0.005)	(0.006)
	[8,342]	[14,563]	[19,259]
50 km	−0.007**	−0.013***	−0.017***
	(0.003)	(0.004)	(0.004)
	[19,422]	[28,483]	[34,256]
Sales		49,873	
Panel C. Modern sales, 2017–2021			
10 km	−0.027*	−0.031*	−0.033*
	(0.016)	(0.019)	(0.019)
	[237]	[508]	[592]
25 km	−0.008	−0.005	−0.008
	(0.014)	(0.012)	(0.013)
	[884]	[1,603]	[1,978]
50 km	−0.010	−0.001	−0.013
	(0.012)	(0.012)	(0.012)
	[2,593]	[3,774]	[4,088]
Sales		6,470	

Notes: All models: dependent variable log price per acre; year and municipality fixed effects; log acreage. Exposure enters as ln(1 + burned area, ha). Each cell is a separate regression. Conley standard errors with a 25 km cutoff in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (Conley). Exposed sales (burned area > 0) in brackets. Source: `output/v4_corrected/sale_level/v4_radius_window_sample_table.csv`.

Table 6. Exposed and Unexposed Sales: Landscape, Soil and Location

	Exposed	Unexposed	Normalized	Within municipality-year	
	mean	mean	difference	Difference	(SE)
Forest share within 10 km	0.444	0.159	1.57	0.075***	(0.010)
Distance to forest (km)	0.625	3.546	−0.86	−0.469***	(0.106)
Grassland/shrubland share within 10 km	0.019	0.022	−0.07	0.004***	(0.001)
Soil capability class (mean; 1 = best)	1.392	2.043	−0.35	−0.053	(0.077)
Dominant soil class 1–3	0.398	0.693	−0.62	−0.099***	(0.026)
Soil class missing	0.664	0.353	0.66	0.053***	(0.020)
Prairies ecozone	0.200	0.769	−1.38	−0.060***	(0.020)
Boreal Plains ecozone	0.449	0.217	0.51	0.008	(0.031)
Boreal Shield ecozone	0.351	0.013	0.97	0.053**	(0.021)
Any fire within 10 km 6–20 years before	0.846	0.240	1.53	0.188***	(0.028)
Log price per acre	5.440	6.165	−0.69	−0.158***	(0.035)
Log acres	4.946	5.101	−0.23	−0.009	(0.015)
Modern file (2017–2021)	0.092	0.117	−0.08	(absorbed)	
Latitude	50.379	50.060	0.37	0.040***	(0.013)
Sales	5,549	50,803			

Notes: Exposed: any burned area within 10 km in the 3 years before the sale. Normalized difference: difference in means divided by the square root of the average of the two variances. Within municipality-year: coefficient on the exposure indicator in a regression of the row variable on the indicator with municipality and year fixed effects; standard errors clustered by municipality; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Soil capability: Canada Land Inventory class (1 = best); dominant class 1–3 marks land with no more than moderate limitations; the class is missing where the CLI does not map the sale point. Ecozones from the National Ecological Framework. Land-cover variables are measured after the sale for sales before 2000. Source: `output/v4_corrected/phase5/p5_balance.csv`.

Table 7. What Explains the Gradient: Stepwise Controls

	(1)	(2)	(3)	(4)	(5)
Panel A. Burned area within 10 km, 3 years before sale					
Coefficient	−0.036***	−0.020***	−0.007	−0.008	−0.012**
	(0.009)	(0.006)	(0.005)	(0.005)	(0.006)
Exposed sales	5,549	5,549	5,549	5,549	2,410
Sales	56,346	56,346	56,346	56,346	32,158
Panel B. Burned area within 25 km, 3 years before sale					
Coefficient	−0.020***	−0.007**	−0.002	−0.002	0.001
	(0.005)	(0.003)	(0.003)	(0.003)	(0.004)
Exposed sales	16,166	16,166	16,166	16,166	7,711
Sales	56,346	56,346	56,346	56,346	32,158
Panel C. Burned area within 50 km, 3 years before sale					
Coefficient	−0.014***	−0.007***	−0.005**		
	(0.004)	(0.002)	(0.002)		
Exposed sales	32,257	32,257	32,257		
Sales	56,346	56,346	56,346		
Year and municipality FE	Yes	Yes	Yes	Yes	Yes
Soil capability and ecoregion		Yes	Yes	Yes	Yes
Land cover			Yes	Yes	Yes
Burned area 6–20 years before				Yes	Yes
Land cover measured before sale only					Yes

Notes: All models: dependent variable log price per acre; year and municipality fixed effects; log acreage. Exposure enters as ln(1 + burned area, ha). Soil capability: Canada Land Inventory dominant class (fixed effects), area-weighted mean class and a missing indicator. Ecoregion: National Ecological Framework ecoregion fixed effects. Land cover: forest share and grassland/shrubland share within 10 km and ln(1 + distance to forest, km) (AAFC Land Use). Column (4) adds ln(1 + burned area 6–20 years before the sale) at the same radius (not constructed for 50 km). Column (5) restricts column (4) to sales in 2000 or later, whose land-cover map predates the sale. Conley standard errors with a 25 km cutoff in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (Conley). Source: `output/v4_corrected/phase5/p5_stepwise.csv`.

Table 8. Lead Placebo: Fire After the Sale

	All sales			All sales, controls			From 2000, controls		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A. Burned area within 10 km									
Before sale	−0.036***	−0.030***		−0.007	−0.005		−0.012**	−0.011**	
	(0.009)	(0.008)		(0.004)	(0.004)		(0.006)	(0.005)	
After sale (lead)		−0.035***	−0.040***		−0.013**	−0.014**		−0.009	−0.010
		(0.008)	(0.009)		(0.006)	(0.006)		(0.006)	(0.006)
Exposed, before / after			5,272 / 5,016					2,133 / 2,112	
Sales			53,025					28,837	
Panel B. Burned area within 25 km									
Before sale	−0.019***	−0.016***		−0.001	0.000		0.002	0.002	
	(0.005)	(0.005)		(0.003)	(0.003)		(0.004)	(0.004)	
After sale (lead)		−0.020***	−0.023***		−0.007**	−0.007**		−0.001	0.000
		(0.005)	(0.005)		(0.003)	(0.003)		(0.004)	(0.004)
Exposed, before / after			15,323 / 14,404					6,868 / 6,823	
Sales			53,025					28,837	

Notes: All models: dependent variable log price per acre; year and municipality fixed effects; log acreage. Exposure enters as ln(1 + burned area, ha). Sample: sales on or before 31 December 2018, for which the 3-year post-sale (lead) window is fully observed in the fire data (records end in 2021); columns (7)–(9) further restrict it to sales from 2000 onward, whose land-cover map predates the sale. Lead exposure: burned area of fires that started after the sale date and within 3 years of it. Controls in columns (4)–(9): land cover, soil capability and ecoregion as in Table 7. Conley standard errors with a 25 km cutoff in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (Conley). Sources: `output/v4_corrected/step3/s3_models.csv`, `s3_lead_placebo_controls.csv` and `phase6/p6_lead_post2000.csv`.

Table 9. Long-Run Fire History

	10 km		25 km	
	(1)	(2)	(3)	(4)
Burned area, 3 years before sale	−0.025***		−0.016***	
	(0.007)		(0.005)	
Burned area, 6–20 years before sale	−0.037***	−0.040***	−0.022***	−0.023***
	(0.008)	(0.008)	(0.006)	(0.007)
Exposed sales, 3 years before	5,549		16,166	
Exposed sales, 6–20 years before	16,906	16,906	35,724	35,724
Sales	56,346	56,346	56,346	56,346

Notes: All models: dependent variable log price per acre; year and municipality fixed effects; log acreage. Exposure enters as ln(1 + burned area, ha). Burned area 6–20 years before the sale counts fires whose date (out date where known, otherwise start date) falls at least 6 and less than 20 years before the sale. Conley standard errors with a 25 km cutoff in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (Conley). Source: `output/v4_corrected/step3/s3_models.csv`.

Table 10. Burned Area within 10 km by Cause and Fire Size (Descriptive)

	Separate models		Joint model		Exposed
Fire class	Coefficient	(SE)	Coefficient	(SE)	sales
Lightning, <10 ha	0.000	(0.044)	0.019	(0.043)	899
Lightning, 10–200 ha	−0.016	(0.019)	−0.009	(0.018)	325
Lightning, ≥200 ha	−0.036*	(0.021)	−0.021	(0.023)	65
Human, <10 ha	−0.060**	(0.025)	−0.044**	(0.021)	3,504
Human, 10–200 ha	−0.022**	(0.009)	−0.011	(0.008)	2,276
Human, ≥200 ha	−0.035***	(0.011)	−0.031***	(0.011)	1,013
Sales			56,346		

Notes: All models: dependent variable log price per acre; year and municipality fixed effects; log acreage. Exposure enters as ln(1 + burned area, ha). Burned area within 10 km in the 3 years before the sale, split by recorded cause (human or lightning) and fire size. Separate models enter one class at a time; the joint model enters all six. The table is descriptive: cause is recorded only as human, lightning or unknown, so agricultural burning cannot be separated from other human-caused fires, and the lightning classes rest on few exposed sales. Conley standard errors with a 25 km cutoff in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (Conley). Source: `output/v4_corrected/step2/s2_models.csv`.

Table 11. Stacked Event Study: Estimates, Pre-Trends and Power

	Unweighted	Wing et al. weights
Panel A. Event-time coefficients (reference: $k = -1$)		
$k = -3$	0.181**	0.180**
	(0.078)	(0.078)
$k = -2$	−0.017	−0.015
	(0.098)	(0.098)
$k = 0$	0.008	0.008
	(0.096)	(0.096)
$k = +1$	0.016	0.015
	(0.103)	(0.103)
$k = +2$	0.017	0.018
	(0.097)	(0.097)
$k = +3$	0.076	0.078
	(0.103)	(0.101)
Panel B. Pooled difference-in-differences and diagnostics		
Pooled post-fire estimate	−0.029	−0.029
	(0.066)	(0.066)
95% confidence interval	[−0.157, 0.100]	[−0.158, 0.100]
Average of $k = 0$ to $+3$ coefficients	0.029	0.030
	(0.062)	(0.061)
	[−0.092, 0.151]	[−0.091, 0.150]
Pooled estimate without $k = -3$	0.029	0.029
	(0.069)	(0.069)
	[−0.107, 0.165]	[−0.107, 0.164]
Minimum detectable effect (80% power)	0.184	0.184
Joint pre-trend test, F (p)	3.59 (0.038)	3.52 (0.040)
Stacks	38	38
Treated sales	527	527
Sales in stacked sample	10,944	
Panel C. Cross-section with the event study's treatment definition		
	Estimate	Exposed sales
≥500 ha within 25 km, prior 3 years	−0.108*** (0.033)	3,460
+ soil capability and ecoregion	−0.045* (0.025)	3,460

Notes: Preferred refined stacked design: 10 km hexagonal cells; a cell is treated in the first year in which burned area within 25 km of its centroid reaches 500 ha; controls are not-yet-treated cells in a 25–50 km ring; window −3 to +3; support requires sales before and after the event in treated and control cells. Fixed effects: stack-specific cell and year (which absorb the match tier); log acreage. Panels A–B: standard errors two-way clustered by cell and stack; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; 95% intervals in brackets. The pooled estimate compares post periods with all pre periods ($k = -3$ to -1); the average of the $k = 0$ to $+3$ coefficients compares them with $k = -1$ only; the pooled estimate without $k = -3$ drops sales at $k = -3$ from every stack. Wing et al. weights follow Wing et al. (2024). The minimum detectable effect is 2.8× the standard error of the pooled estimate (5% two-sided test, 80% power). Panel C: sale-level model of Table 5 with an indicator that annual burned area within 25 km reached 500 ha in the 3 years before the sale; Conley (25 km) standard errors in parentheses. Sources: `output/v4_corrected/step3/event_pretrends/` and `s3_item5_comparison.csv`.

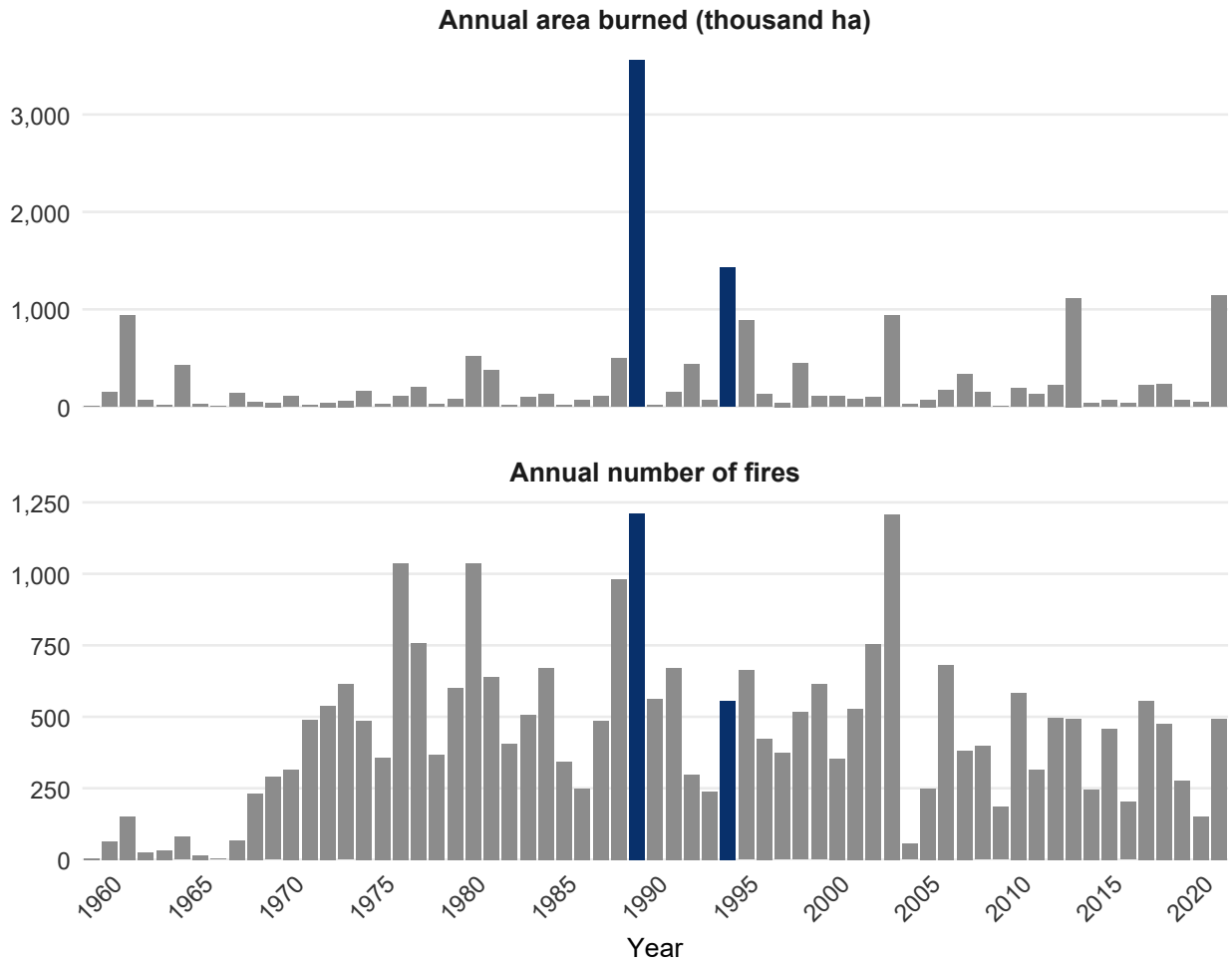


Figure 1. Annual Area Burned and Annual Number of Fires in Manitoba, 1959–2021

Notes: Top panel: total annual area burned (thousand hectares); bottom panel: annual number of recorded fires. Fire records reported by the Manitoba agency in the Canadian National Fire Database, which begin in 1959. Dark bars mark 1989 and 1994, the two largest burned-area years.

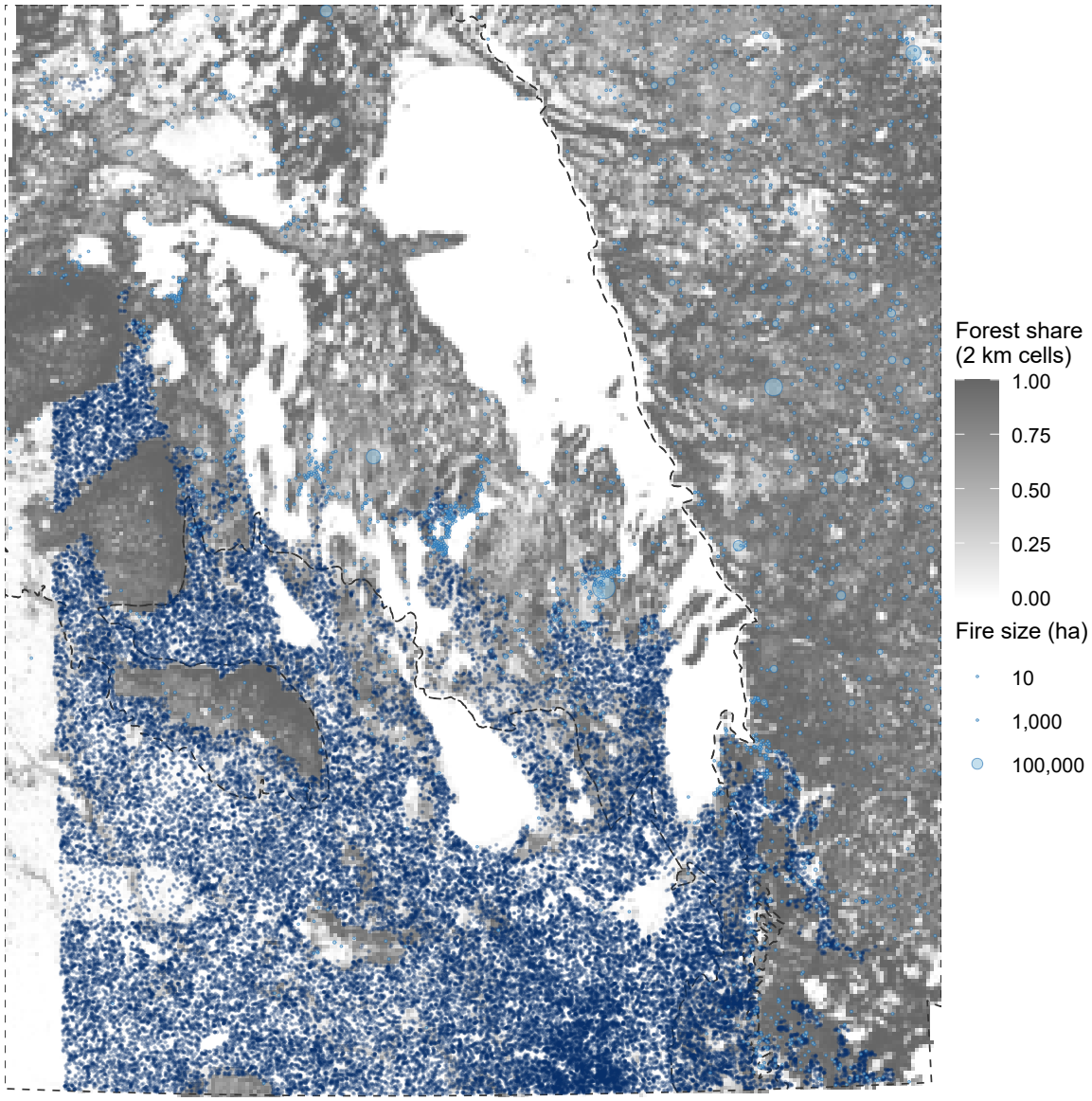


Figure 2. Farmland Sales, Fires and Forest Cover

Notes: Dark blue points: the 56,352 sales in the farmland sample, 1987–2021. Light blue circles: Canadian National Fire Database fires of at least 10 ha with fire dates from 1982 to 2021, sized by burned area (prescribed burns excluded). Grey shading: forest share of 2 km cells from the AAFC Land Use 2010 map (white: no forest, including cropland and water). Dashed lines: ecozone boundaries (National Ecological Framework).

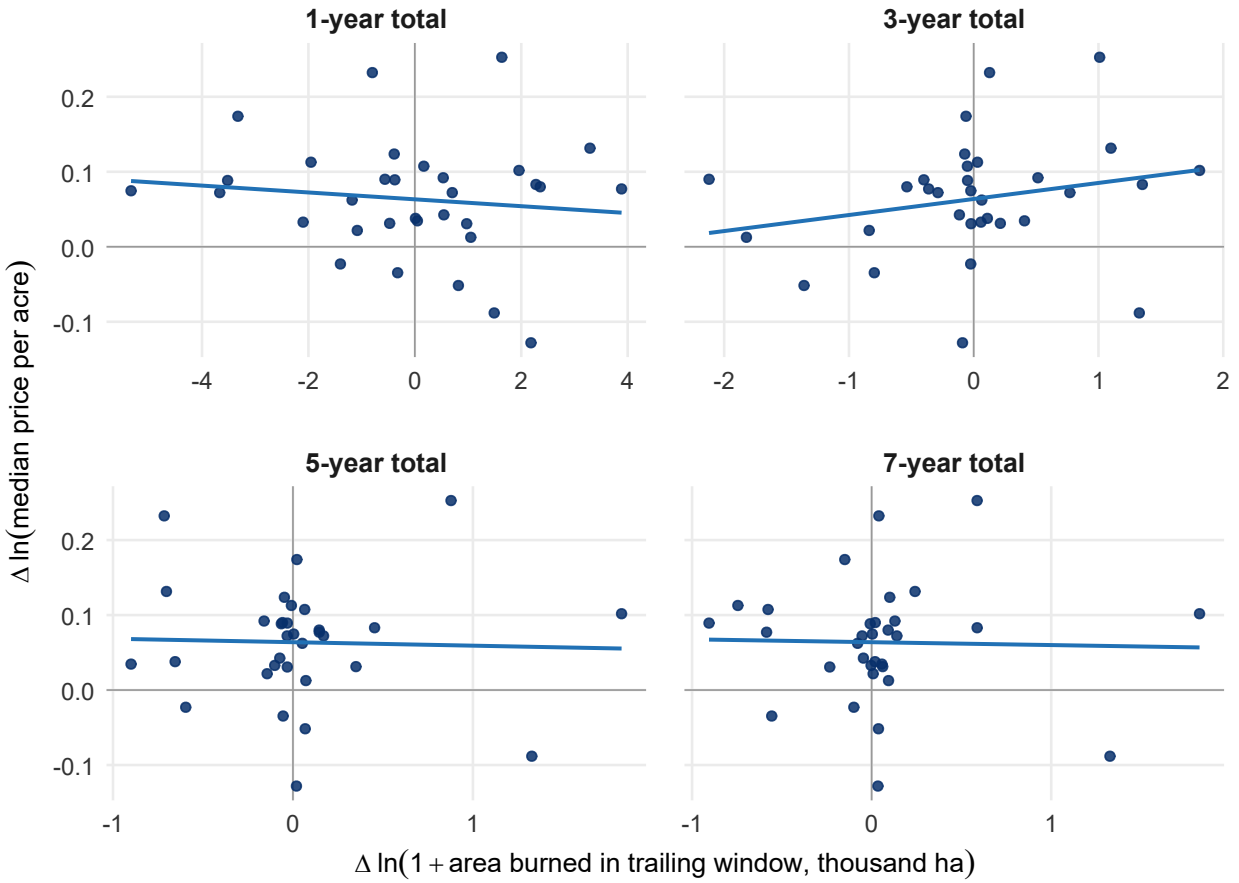


Figure 3. Annual Changes in Median Price per Acre and Area Burned

Notes: Each point is a year from 1988 to 2020. Vertical axis: change in the log of the annual median price per acre in the farmland sample; horizontal axis: change in the log of one plus Manitoba area burned (thousand hectares) over the trailing 1-, 3-, 5- or 7-year window. The change from 2016 to 2017, which coincides with the change of sales file, is excluded. Lines are linear fits.

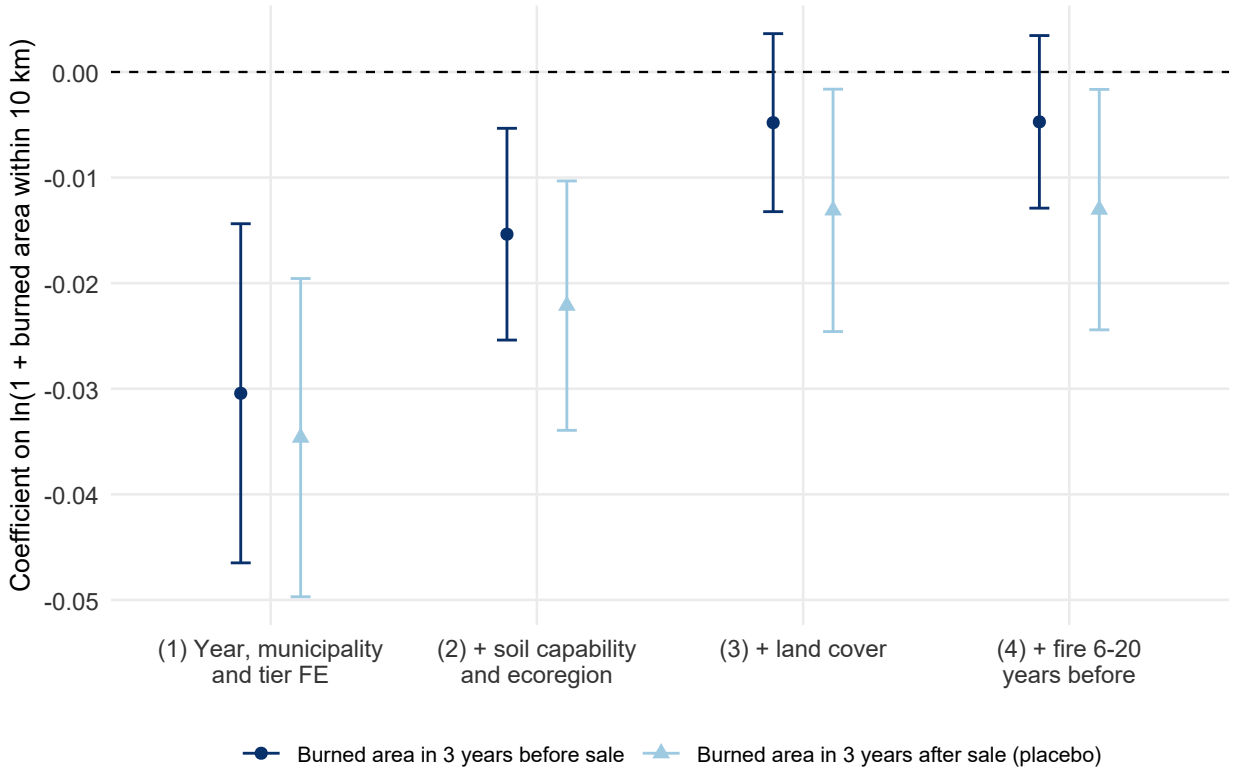


Figure 4. Burned Area Before and After the Sale across Specifications

Notes: Coefficients on ln(1 + burned area within 10 km) in the 3 years before the sale (dark circles) and in the 3 years after the sale (light triangles), estimated jointly (equation 6) on the 53,025 sales whose post-sale window is fully observed. Specifications (1)–(4) add controls cumulatively as in Table 7. Bars: 95% confidence intervals from Conley standard errors (25 km cutoff).

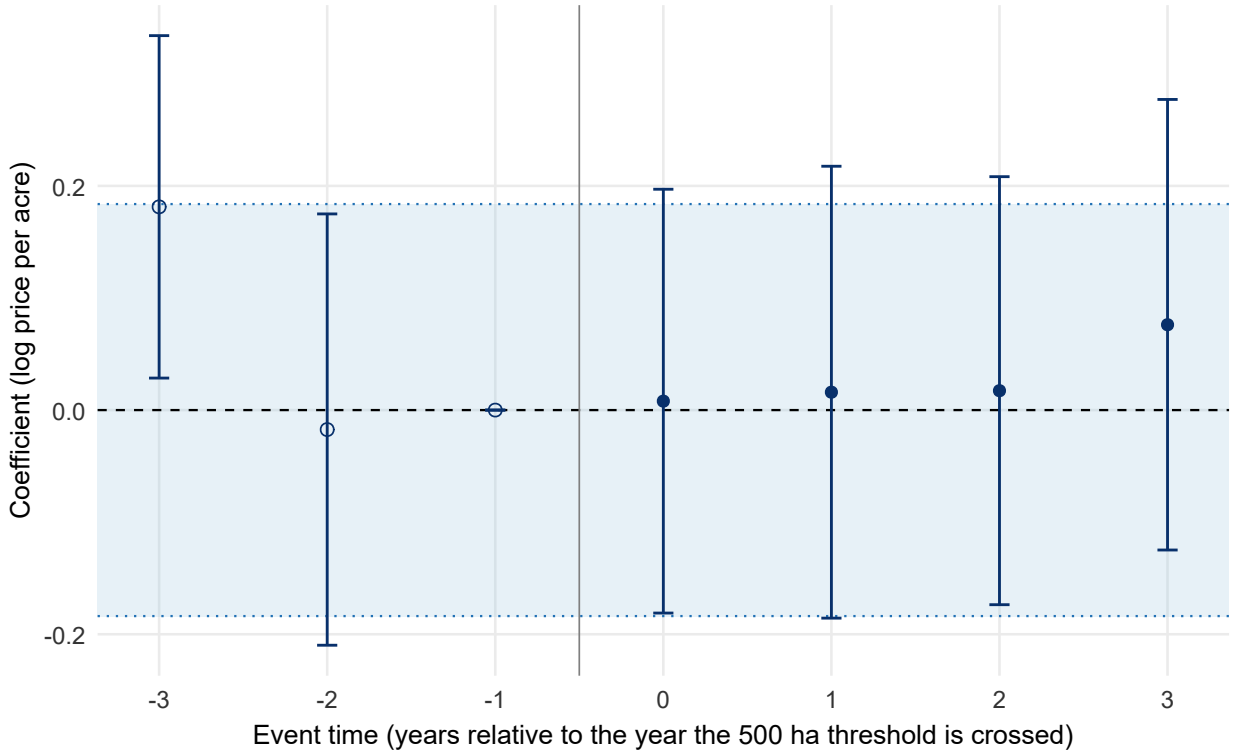


Figure 5. Stacked Event Study around Large Nearby Fires

Notes: Event-time coefficients β_k from equation (7) with 95% confidence intervals (two-way clustered by cell and stack), unweighted preferred design (38 stacks, 527 treated sales); $k = -1$ is the reference period. Open circles: pre-period; filled circles: post-period. The shaded band and dotted lines mark $\pm$ the minimum detectable effect of the pooled estimate at 80% power (0.184 log points). See Table 11.

Appendix: Additional Results

Table A1. Alternative Sample Definitions

	(1) Main sample	(2) With buildings	(3) 40+ acres	(4) Un-trimmed	(5) Bare land confirmed
10 km, 1 yr	−0.029***	−0.023***	−0.029***	−0.034***	−0.029***
	(0.009)	(0.008)	(0.009)	(0.010)	(0.009)
	[2,681]	[3,597]	[2,562]	[2,749]	[2,670]
10 km, 3 yr	−0.036***	−0.032***	−0.036***	−0.038***	−0.036***
	(0.009)	(0.008)	(0.009)	(0.010)	(0.009)
	[5,549]	[7,443]	[5,312]	[5,679]	[5,521]
10 km, 5 yr	−0.038***	−0.034***	−0.038***	−0.041***	−0.038***
	(0.010)	(0.009)	(0.010)	(0.011)	(0.010)
	[7,860]	[10,546]	[7,546]	[8,049]	[7,829]
25 km, 1 yr	−0.015***	−0.014***	−0.015***	−0.017***	
	(0.005)	(0.004)	(0.005)	(0.005)	
	[9,226]	[12,191]	[8,880]	[9,440]	
25 km, 3 yr	−0.020***	−0.019***	−0.019***	−0.021***	
	(0.005)	(0.004)	(0.005)	(0.005)	
	[16,166]	[21,342]	[15,611]	[16,534]	
25 km, 5 yr	−0.021***	−0.019***	−0.021***	−0.022***	
	(0.006)	(0.005)	(0.006)	(0.006)	
	[21,237]	[28,132]	[20,543]	[21,725]	
Sales	56,346	72,935	54,448	57,527	55,909

Notes: All models: dependent variable log price per acre; year and municipality fixed effects; log acreage. Exposure enters as ln(1 + burned area, ha). (1) Main sample. (2) Adds historical sales classed as farmland with buildings. (3) Requires at least 40 acres. (4) No trimming of price per acre. (5) Main sample excluding the modern sales whose assessment reports show buildings, a residential portion or a non-farm class at the sale's tax year (reports were retrieved for about half of modern sales; unclassified sales are kept). Conley standard errors with a 25 km cutoff in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (Conley). Exposed sales in brackets. Sources: `output/v4_corrected/sale_level/v4_variant_table.csv`, `step3/s3_item8_rerun_models.csv`.

Table A2. Burned Area and Log Price per Acre: Municipality-Clustered Standard Errors

	Trailing window		
Radius	1 year	3 years	5 years
Panel A. All sales, 1987–2021			
10 km	−0.029***	−0.036***	−0.038***
	(0.008)	(0.008)	(0.008)
	[2,681]	[5,549]	[7,860]
25 km	−0.015***	−0.020***	−0.021***
	(0.004)	(0.005)	(0.005)
	[9,226]	[16,166]	[21,237]
50 km	−0.008**	−0.014***	−0.018***
	(0.003)	(0.004)	(0.005)
	[22,015]	[32,257]	[38,344]
Sales		56,346	
Panel B. Historical sales, 1987–2016			
10 km	−0.028***	−0.036***	−0.039***
	(0.008)	(0.009)	(0.008)
	[2,444]	[5,041]	[7,268]
25 km	−0.015***	−0.020***	−0.021***
	(0.004)	(0.005)	(0.005)
	[8,342]	[14,563]	[19,259]
50 km	−0.007**	−0.013***	−0.017***
	(0.003)	(0.004)	(0.005)
	[19,422]	[28,483]	[34,256]
Sales		49,873	
Panel C. Modern sales, 2017–2021			
10 km	−0.027*	−0.031**	−0.033*
	(0.014)	(0.015)	(0.019)
	[237]	[508]	[592]
25 km	−0.008	−0.005	−0.008
	(0.011)	(0.010)	(0.013)
	[884]	[1,603]	[1,978]
50 km	−0.010	−0.001	−0.013
	(0.011)	(0.013)	(0.012)
	[2,593]	[3,774]	[4,088]
Sales		6,470	

Notes: All models: dependent variable log price per acre; year and municipality fixed effects; log acreage. Exposure enters as ln(1 + burned area, ha). Each cell is a separate regression. Standard errors clustered by municipality in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (clustered). Exposed sales (burned area > 0) in brackets. Source: `output/v4_corrected/sale_level/v4_radius_window_sample_table.csv`.

Table A3. Stepwise Controls: Municipality-Clustered Standard Errors

	(1)	(2)	(3)	(4)	(5)
Panel A. Burned area within 10 km, 3 years before sale					
Coefficient	−0.036***	−0.020***	−0.007	−0.008	−0.012**
	(0.008)	(0.006)	(0.005)	(0.005)	(0.006)
Exposed sales	5,549	5,549	5,549	5,549	2,410
Sales	56,346	56,346	56,346	56,346	32,158
Panel B. Burned area within 25 km, 3 years before sale					
Coefficient	−0.020***	−0.007**	−0.002	−0.002	0.001
	(0.005)	(0.003)	(0.003)	(0.003)	(0.004)
Exposed sales	16,166	16,166	16,166	16,166	7,711
Sales	56,346	56,346	56,346	56,346	32,158
Panel C. Burned area within 50 km, 3 years before sale					
Coefficient	−0.014***	−0.007***	−0.005**		
	(0.004)	(0.002)	(0.002)		
Exposed sales	32,257	32,257	32,257		
Sales	56,346	56,346	56,346		
Year and municipality FE	Yes	Yes	Yes	Yes	Yes
Soil capability and ecoregion		Yes	Yes	Yes	Yes
Land cover			Yes	Yes	Yes
Burned area 6–20 years before				Yes	Yes
Land cover measured before sale only					Yes

Notes: All models: dependent variable log price per acre; year and municipality fixed effects; log acreage. Exposure enters as ln(1 + burned area, ha). Soil capability: Canada Land Inventory dominant class (fixed effects), area-weighted mean class and a missing indicator. Ecoregion: National Ecological Framework ecoregion fixed effects. Land cover: forest share and grassland/shrubland share within 10 km and ln(1 + distance to forest, km) (AAFC Land Use). Column (4) adds ln(1 + burned area 6–20 years before the sale) at the same radius (not constructed for 50 km). Column (5) restricts column (4) to sales in 2000 or later, whose land-cover map predates the sale. Standard errors clustered by municipality in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (clustered). Source: `output/v4_corrected/phase5/p5_stepwise.csv`.

Table A4. Lead Placebo: Municipality-Clustered Standard Errors

	All sales			All sales, controls			From 2000, controls		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A. Burned area within 10 km									
Before sale	−0.036***	−0.030***		−0.007	−0.005		−0.012*	−0.011*	
	(0.008)	(0.007)		(0.004)	(0.004)		(0.006)	(0.006)	
After sale (lead)		−0.035***	−0.040***		−0.013*	−0.014**		−0.009	−0.010
		(0.009)	(0.009)		(0.007)	(0.007)		(0.007)	(0.007)
Exposed, before / after			5,272 / 5,016					2,133 / 2,112	
Sales			53,025					28,837	
Panel B. Burned area within 25 km									
Before sale	−0.019***	−0.016***		−0.001	0.000		0.002	0.002	
	(0.005)	(0.004)		(0.003)	(0.003)		(0.004)	(0.004)	
After sale (lead)		−0.020***	−0.023***		−0.007**	−0.007*		−0.001	0.000
		(0.005)	(0.006)		(0.003)	(0.003)		(0.004)	(0.004)
Exposed, before / after			15,323 / 14,404					6,868 / 6,823	
Sales			53,025					28,837	

Notes: All models: dependent variable log price per acre; year and municipality fixed effects; log acreage. Exposure enters as ln(1 + burned area, ha). Sample: sales on or before 31 December 2018, for which the 3-year post-sale (lead) window is fully observed in the fire data (records end in 2021); columns (7)–(9) further restrict it to sales from 2000 onward, whose land-cover map predates the sale. Lead exposure: burned area of fires that started after the sale date and within 3 years of it. Controls in columns (4)–(9): land cover, soil capability and ecoregion as in Table 7. Standard errors clustered by municipality in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (clustered). Sources: `output/v4_corrected/step3/s3_models.csv`, `s3_lead_placebo_controls.csv` and `phase6/p6_lead_post2000.csv`.

Table A5. Lead Placebo by Cause and Distance (10 km)

	Without controls		With land cover, soil, ecoregion		Exposed
Lead (3 years after sale) restricted to	Lead	Before sale	Lead	Before sale	sales (lead)
Lightning-caused fires	0.001	−0.036***	0.006	−0.007	1,007
	(0.011)	(0.009)	(0.007)	(0.004)	
Human-caused fires	−0.035***	−0.031***	−0.013**	−0.005	4,624
	(0.008)	(0.008)	(0.006)	(0.004)	
Fires more than 2 km from the sale point	−0.033***	−0.031***	−0.013**	−0.005	4,913
	(0.008)	(0.008)	(0.006)	(0.004)	
Exposed sales, before sale; sales			5,272; 53,025		
Lightning-caused fires only, before and after the sale					
Lightning lead with lightning before	−0.014	−0.025*	0.004	−0.002	1,007
	(0.012)	(0.014)	(0.007)	(0.010)	
Exposed sales, lightning before sale			1,101		
Sales from 2000 onward, with controls (columns 4–5 only)					
Human-caused fires			−0.009	−0.011**	1,986
			(0.006)	(0.005)	
Lightning-caused fires			−0.002	−0.012**	192
			(0.015)	(0.006)	
Sales from 2000 onward			28,837		

Notes: All models: dependent variable log price per acre; year and municipality fixed effects; log acreage. Exposure enters as ln(1 + burned area, ha). Each row in the upper panel is a model with burned area within 10 km in the 3 years before the sale (all fires) and lead burned area within 10 km restricted as stated (fires that started after the sale and within 3 years of it). The middle panel enters lightning-caused burned area before and after the sale together; the last panel restricts to sales from 2000 onward, with controls. Sample: sales on or before 31 December 2018. Controls as in Table 7. Conley standard errors with a 25 km cutoff in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (Conley). Sources: `output/v4_corrected/phase5b/p5b_lead_variants.csv`, `phase6/p6_lightning_pre_lead.csv`.

Table A6. Burned Area from Fire Perimeters (NBAC)

	Trailing window			3 years, with
Radius	1 year	3 years	5 years	controls
10 km	−0.023**	−0.031***	−0.029***	−0.005
	(0.009)	(0.009)	(0.010)	(0.005)
	[1,430]	[3,449]	[4,857]	[3,449]
25 km	−0.011**	−0.013***	−0.010*	0.000
	(0.005)	(0.005)	(0.005)	(0.003)
	[5,617]	[11,332]	[14,923]	[11,332]
Agreement with the fire-point measure (3 years)				
10 km: share exposed, points / perimeters	0.098 / 0.061; correlation of logs 0.64			
25 km: share exposed, points / perimeters	0.287 / 0.201; correlation of logs 0.72			
Sales	56,346			

Notes: All models: dependent variable log price per acre; year and municipality fixed effects; log acreage. Exposure enters as ln(1 + burned area, ha). Burned area is the area of National Burned Area Composite perimeters (Hall et al., 2020) within the radius, rasterized to 250 m cells, for fires that ended strictly before the sale (agency or hotspot end date; otherwise start date by the end of the month before the sale month); prescribed burns excluded. Controls: land cover, soil capability and ecoregion as in Table 7. Conley standard errors with a 25 km cutoff in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (Conley). Exposed sales in brackets. Source: `output/v4_corrected/phase5/p5_nbac_models.csv`.

Table A7. Including Fire Records without a Start Date (Dated 1 July)

	Trailing window		
Radius	1 year	3 years	5 years
Panel A. Gradient, all sales			
10 km	-0.026^{***}	-0.036^{***}	-0.037^{***}
	(0.009)	(0.010)	(0.010)
	[3,095]	[6,451]	[8,928]
25 km	-0.015^{**}	-0.021^{***}	-0.020^{***}
	(0.006)	(0.006)	(0.006)
	[10,887]	[18,847]	[24,031]
50 km	-0.009^{**}	-0.015^{***}	-0.018^{***}
	(0.004)	(0.004)	(0.004)
	[25,347]	[35,986]	[41,377]
Panel B. 10 km, 3 years: controls and lead placebo			
	Coefficient	(SE)	Exposed
+ soil capability and ecoregion	-0.019^{***}	(0.006)	6,451
+ land cover	-0.006	(0.005)	6,451
Before sale (with lead)	-0.030^{***}	(0.008)	6,174
Lead (with before sale)	-0.032^{***}	(0.007)	6,117
Before sale (with lead), controls	-0.004	(0.005)	6,174
Lead (with before sale), controls	-0.010^{**}	(0.005)	6,117

Notes: All models: dependent variable log price per acre; year and municipality fixed effects; log acreage. Exposure enters as ln(1 + burned area, ha). Exposure adds the fire records that have no start date, dated 1 July of their fire year with no out date (prescribed burns excluded). Lead placebo on sales on or before 31 December 2018. Controls as in Table 7. Conley standard errors with a 25 km cutoff in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (Conley). Exposed sales in brackets in Panel A. Sources: `output/v4_corrected/phase6/p6_undated_models.csv`, `fire/undated_fires_summary.csv`.

Table A8. Event-Study Designs, Leave-One-Out and Sensitivity Bounds

Design	Treated units	Treated sales	Pooled estimate	(SE)	p
Panel A. Designs (500 ha within 25 km)					
Refined, 25–50 km ring (preferred)	38 (13; 22)	527	−0.029	(0.066)	0.662
Refined, 50–100 km ring	41 (16; 41)	558	−0.003	(0.063)	0.960
First stacked, 25–75 km ring, ±5 years	92 (21; 72)	1,292	0.019	(0.040)	0.640
Broad, never-treated controls, ±5 years	92 (21; 72)	1,292	0.003	(0.034)	0.918
Panel B. Refined design: threshold and radius grid					
500 ha, 10 km (ring 10–20 km)			0.122	(0.149)	0.453
500 ha, 25 km (ring 25–50 km)			−0.029	(0.066)	0.662
500 ha, 50 km (ring 50–100 km)			0.039	(0.031)	0.206
1,000 ha, 10 km (ring 10–20 km)			0.105	(0.139)	0.476
1,000 ha, 25 km (ring 25–50 km)			0.008	(0.065)	0.898
1,000 ha, 50 km (ring 50–100 km)			0.016	(0.031)	0.616
Leave one event year out (9 distinct runs; omitting 1991 reproduces the baseline): range			[−0.056, 0.007]		min 0.441
Panel C. HonestDiD relative-magnitude bounds, preferred design					
$\bar{M}$	First post period		Average of post periods 0 to +3		
0 (original CI)	[−0.18, 0.20]		[−0.09, 0.15]		
0.5	[−0.27, 0.30]		[−0.43, 0.51]		
1	[−0.40, 0.44]		[−0.87, 0.95]		
1.5	[−0.56, 0.60]		[−1.24, 1.24]		
2	[−0.73, 0.77]		[−1.24, 1.24]		
Panel D. Preferred design: Riding Mountain National Park and prescribed burns					
Sample	Stacks	Treated sales	Pooled estimate	(SE)	Pre-trend p
Prescribed burns removed (main)	38	527	−0.029	(0.066)	0.038
Prescribed burns included	43	600	−0.001	(0.065)	0.029
Excluding stacks within 10 km of the park	30	418	−0.070	(0.081)	0.026
Excluding stacks and control cells within 10 km	30	418	−0.069	(0.081)	0.026
Refined, 50–100 km ring: removed / included	41 / 46	558 / 631	−0.003 / 0.022		
First stacked: removed / included	92 / 114	1,292 / 1,575	0.019 / 0.008		
Broad: removed / included	92 / 114	1,292 / 1,575	0.003 / −0.003		

Notes: Pooled post-fire difference-in-differences estimates. Treated units are stacks (stacked designs) or treated cells (broad design); in parentheses, the number of distinct fires that are the largest contributor to a treated unit's event-year burned area, and the number of distinct fires contributing to it. Standard errors

Table A9. Controls from the 2026 Assessment Roll (Possibly Post-Treatment)

	(1)	(2)	(3)
Burned area within 10 km, 3 years	-0.020^{***}	-0.020^{***}	-0.012^{***}
Conley SE	(0.006)	(0.006)	(0.004)
Municipality-clustered SE	(0.006)	(0.006)	(0.004)
Exposed sales	5,549	5,549	5,549
Sales	56,346	56,346	56,346
Soil capability and ecoregion	Yes	Yes	Yes
Dwellings on the sale's parcels (2026 roll)		Yes	Yes
Log assessed value per acre (2026 roll)			Yes

Notes: All models: dependent variable log price per acre; year and municipality fixed effects; log acreage. Exposure enters as $\ln(1 + \text{burned area, ha})$. The 2026 roll is recorded after almost every sale and the assessed value measures land value itself, so columns (2) and (3) may control for outcomes of fire exposure; they are shown only as a check. Stars use Conley standard errors: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: `output/v4_corrected/step2/s2_models.csv`.

Table A10. Validation of the Parcel Linkage

Test	Truncated roll-number link	Internal-identifier link
Panel A. Agreement between sale and linked parcels (historical sales placed by both links)		
Correlation, log price and log 2026 assessed value	0.317	0.697
Within municipality-year R^2, log price on log assessed value	0.011	0.537
Within municipality-year R^2, log price on log linked acreage	0.021	0.333
Bare-land sales whose linked parcels have a dwelling (2026)	0.127	0.094
Panel B. Share of historical identifiers matching an internal identifier		
Same municipality	0.992	
Next municipality number (placebo)	0.000	
Identifier + 7, same municipality (placebo)	0.884	
Identifier + 1000, same municipality (placebo)	0.472	
Panel C. Linkage of historical sales, 1984–2016		
Linked on all parcels	386,794 (99.17%)	
Linked on some parcels	942 (0.24%)	
Not linked	2,310 (0.59%)	

Notes: Panel A compares, for the historical sales placed by both methods, the parcels assigned by roll number (truncated to the nearest hundred and matched only where that key identifies a single parcel) with those assigned by the assessment system's internal roll identifier. Panel B: identifiers are allocated in dense blocks within municipality, so the placebo rows show how often an identifier matches by chance; the value and acreage tests in Panel A, not the match rate, establish that the linked parcel is the sale's parcel. Sources: `output/v3_relinked/step0/linkage_tests.csv` (historical sales, unchanged in v4), `sample/v4_linkage_summary.csv`.

Table A11. Multiple Testing

Panel A. Cross-sectional exposure coefficients estimated Set	Coefficients
Gradient: 3 radii x 3 windows x 3 samples (full, historical, modern)	27
Main sale-level table: 25 km burned area and counts x 3 windows x 3 FE sets	18
Sample definitions: 10/25 km area and counts x 3 windows x 5 samples	60
Fire cause, size class, soil and checks (Step 2)	93
Lead placebo, history, land cover, event definition (Step 3)	28
Lead placebo with controls	8
Excluding modern sales with buildings	15
Stepwise specifications	13
Stepwise lead placebo	8
NBAC perimeters	8
Lead placebo by cause and distance	12
Lightning-caused fires before and after the sale	8
Undated fires dated 1 July	15
Lead placebo, sales from 2000 onward	12
Total	325

Panel B. Holm-adjusted p-values, main gradient (all sales, Conley) Radius, window	Unadjusted p	Holm-adjusted p
10 km, 1 yr	0.001	0.003
10 km, 3 yr	<0.001	<0.001
10 km, 5 yr	<0.001	<0.001
25 km, 1 yr	0.003	0.006
25 km, 3 yr	<0.001	<0.001
25 km, 5 yr	<0.001	0.002
50 km, 1 yr	0.011	0.011
50 km, 3 yr	<0.001	<0.001
50 km, 5 yr	<0.001	<0.001

Notes: Panel A counts each distinct exposure coefficient once (not once per standard-error type) across the v3 sale-level models, including radius, window, sample, fire-class and specification variants; event-study designs are not included. Panel B applies the Holm (1979) step-down adjustment to the nine radius-by-window coefficients of Table 5, Panel A. Sources: the CSV files listed in Panel A under `output/v4_corrected/`.